

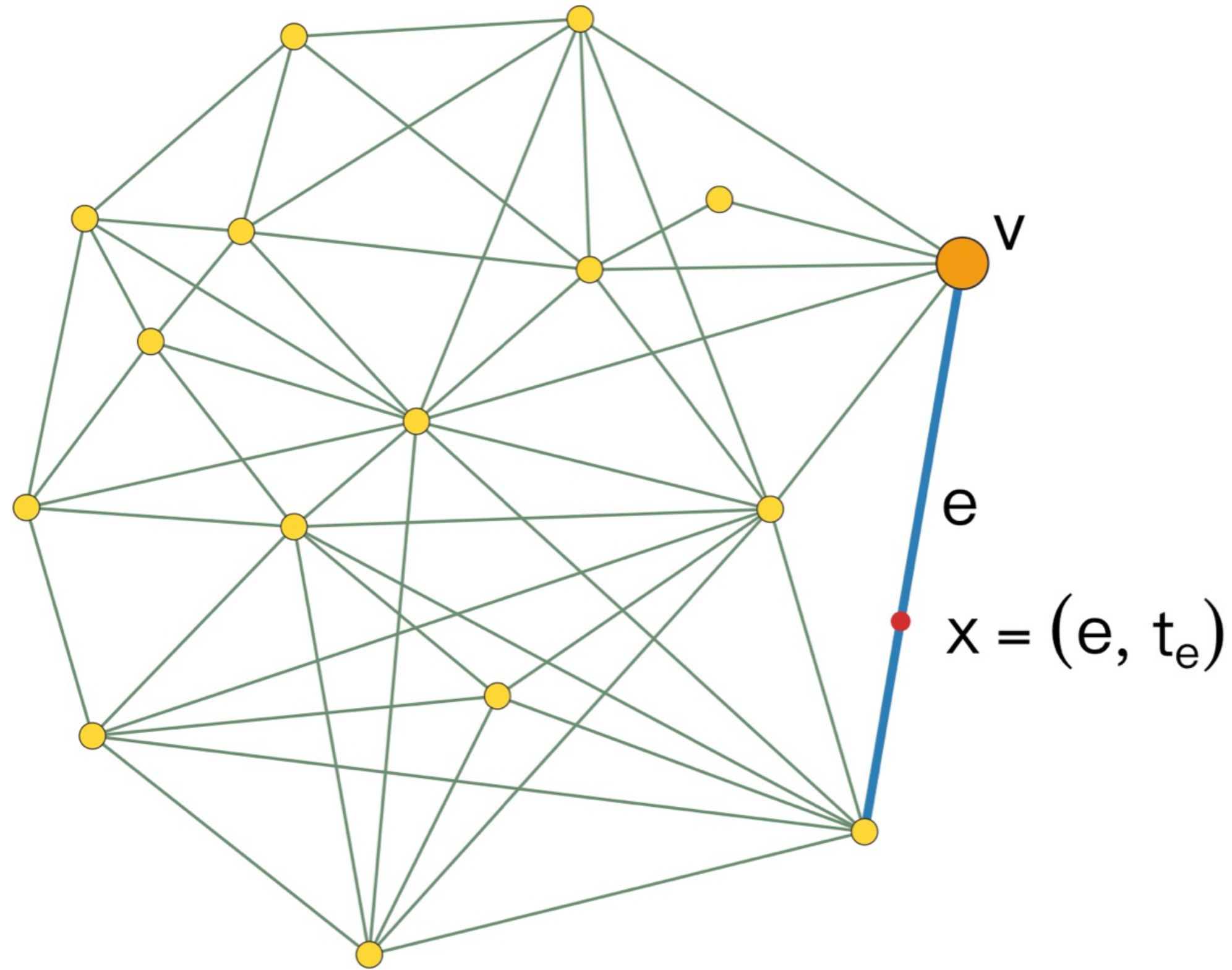
Bayesian Optimization on Networks

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Joint with Daniel Sanz-Alonso + Ruiyi Yang

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What Is a “Metric Graph” ?



New York Telecom Network



Mouse Cortical Neuron

1. Finite vertices/edges

2. Finite length

$$(e \cong [0, L_e])$$

3. Intrinsic metric:

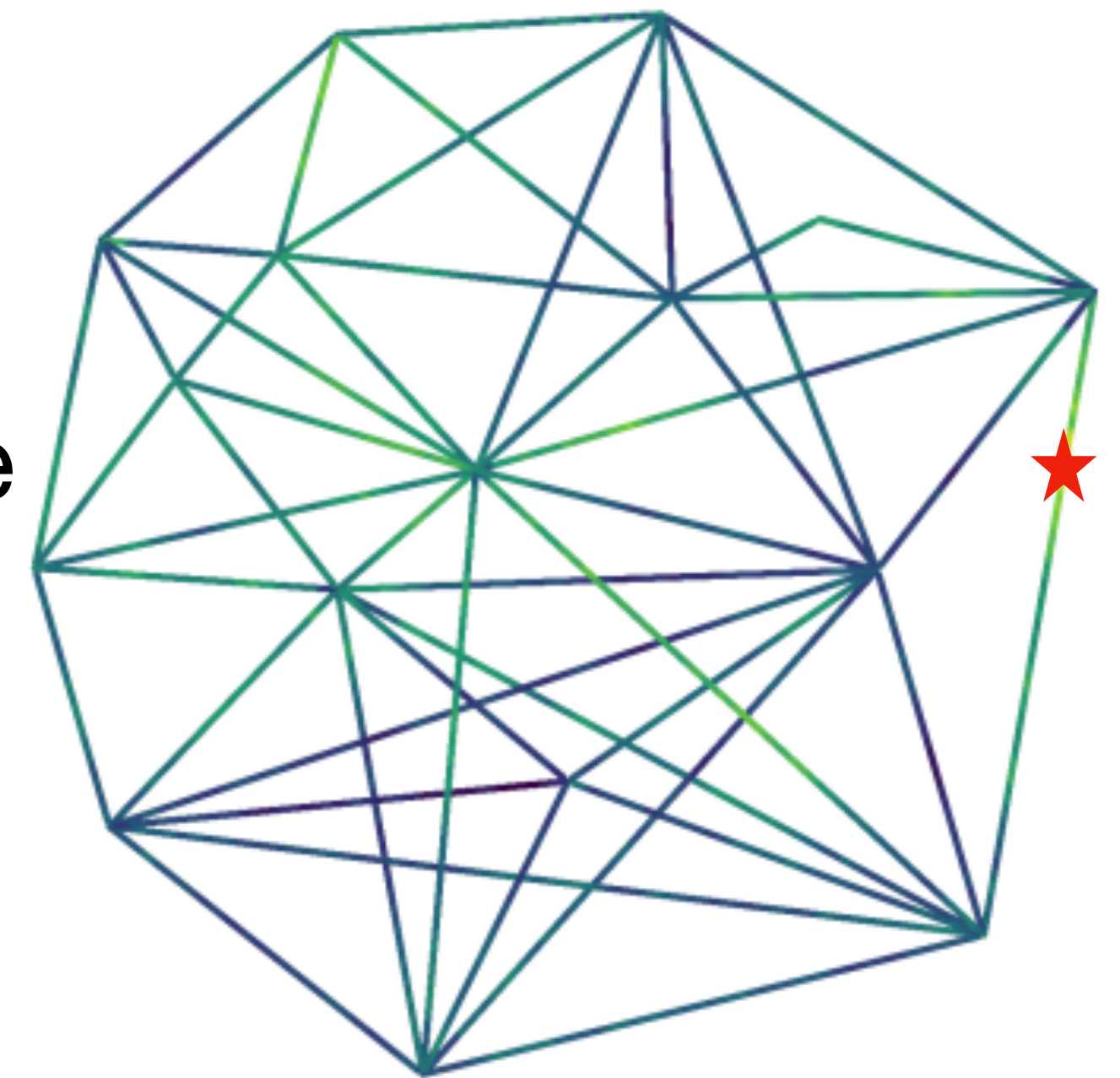
Shortest-path distance

Resistance distance

Goal: Find global maximizer of a “black-box” objective $f^\dagger : \Gamma \rightarrow \mathbb{R}$

Motivating setting:

- High fidelity observation of $f^\dagger$ are costly to obtain,
- Only noisy pointwise queries $y = f^\dagger(x) + \epsilon$ are available
- Reliable derivatives of $f^\dagger$ are unavailable
- $f^\dagger$ “smooth” on the metric graph

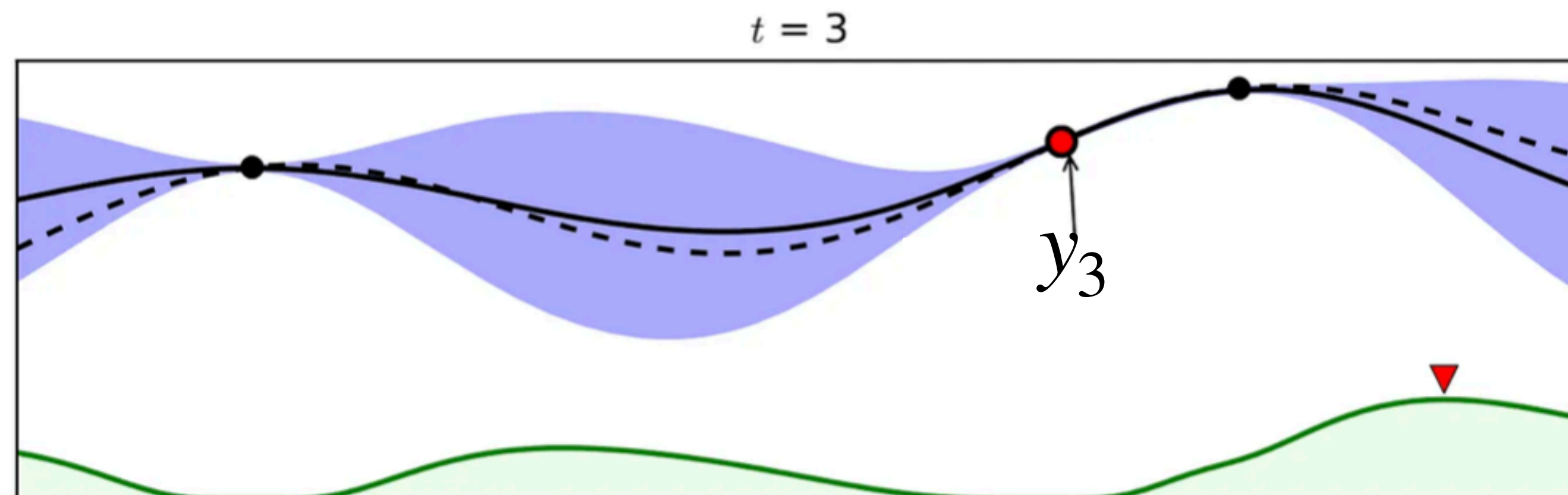
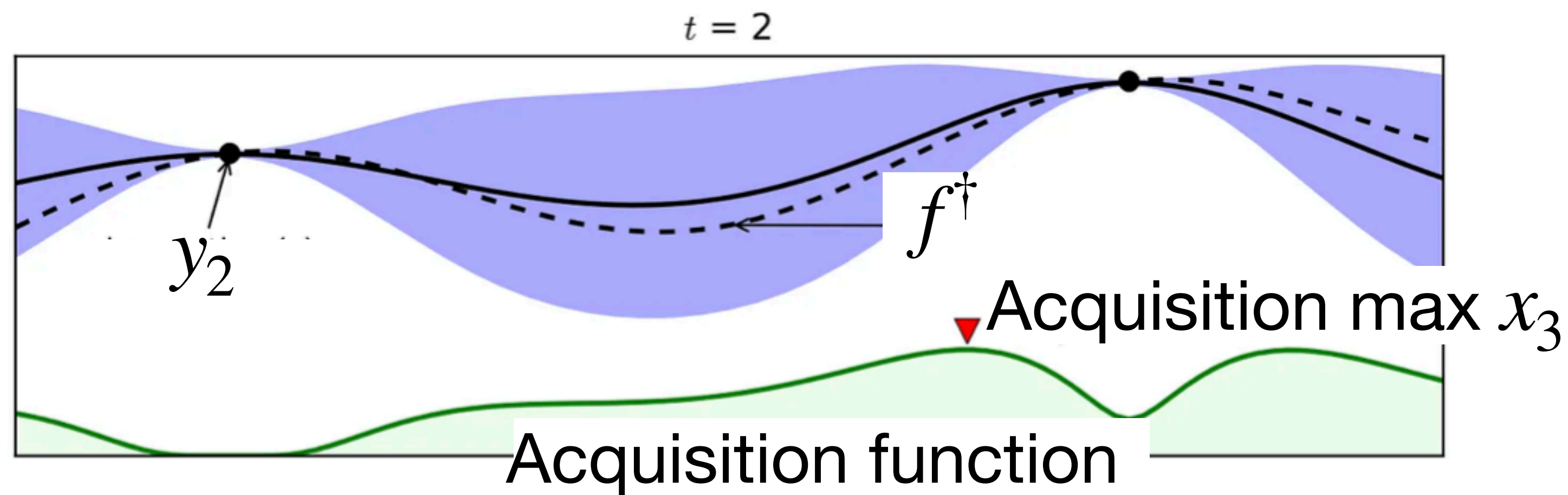


Contribution: BayesOpt on networks with convergence guarantees

Outline

- **What is BayesOpt?**
- What ingredients matter for good performance of BayesOpt?
- BayesOpt on metric graphs and regret bounds
- Numerical examples

- BayesOpt:**
- Build surrogate model for $f^\dagger$ using past data $\{(x_t, y_t)\}$
 - Use surrogate to define an acquisition function
 - Obtain next query x_t by maximizing acquisition function



Dashed: Objective
Solid: surrogate mean
Shaded: surrogate UQ

Acquisition Functions

Given $\mathcal{D}_{t-1}$ and a GP prior $f^\dagger \sim GP(0, k)$ we obtain **surrogate model**:

$$\mu_{t-1}(x) := k_{t-1}(x)^\top (K_{t-1} + \lambda I)^{-1} Y_{t-1}, \quad \text{Mean}$$

$$k_{t-1}(x, x') := k(x, x') - k_{t-1}(x)^\top (K_{t-1} + \lambda I)^{-1} k_{t-1}(x'), \quad \text{Covariance function}$$

$$\sigma_{t-1}^2(x) := k_{t-1}(x, x), \quad \text{Marginal variance}$$

We then consider two **acquisition functions**:

$$\begin{cases} \mu_{t-1}(x) + \beta_t \sigma_{t-1}(x) & \text{(IGP-UCB)} \\ f_t(x), \quad f_t \sim GP(\mu_{t-1}, v_t^2 k_{t-1}) & \text{(GP-TS)} \end{cases}$$

Here, β_t, v_t are hyperparameters to balance exploration vs exploitation

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Ingredients that matter for good BayesOpt

A suitable surrogate model :

The GP prior encodes prior belief about $f^\dagger$. Together with data, it yields a posterior surrogate that guides query selection.

Geometry-aware kernels:

Classical Euclidean kernels may fail to capture the intrinsic geometry of metric graphs.

Robustness to misspecification:

The regret guarantees should remain meaningful when the kernel smoothness does not match the true regularity of $f^\dagger$.

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Choice of kernel in Metric Graph Setting

In $\mathbb{R}^d$, a Matérn field u is the (weak) solution to the fractional SPDE:

$$\tau (\kappa^2 - \Delta)^{\alpha/2} u = \mathcal{W}, \quad \nu = \alpha - \frac{d}{2} \quad [\text{Lindgren-Rue-Lindström, 2011}]$$

Classical Euclidean kernels are built from Euclidean geometry, and are **not necessarily compatible** with shortest-path geometry on a metric graph.

On a metric graph Γ , can we follow the same fractional SPDE principle to derive a more natural geometry-aware kernel?

Function Spaces & Differential Operators on Γ

Edge parameterization ($e \cong [0, L_e]$) $\rightarrow$ **edgewise definition**

$(f|_e \in L^2(e), H^1(e), \dots) \rightarrow$ **coupling via vertex conditions**

$$f = \bigoplus_{e \in E} f|_e, \quad L^2(\Gamma) = \bigoplus_{e \in E} L^2(e), \quad H^1(\Gamma) = \left\{ p : p|_e \in H^1(e) \text{ and } p \text{ is globally continuous} \right\}.$$

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$$(\mathcal{L}p)_e(t) = -\frac{d}{dt} \left(a_e(t) \frac{d}{dt} p_e(t) \right) + \kappa_e^2(t) p_e(t), \quad t \in (0, L_e).$$

Standard Kirchhoff vertex conditions:

$$\begin{cases} p \text{ is continuous on } \Gamma, \\ \sum_{e \in E_v} a_e(v) \partial_e p(v) = 0 \quad \forall v \in V. \end{cases}$$

Differential operator $\mathcal{L}$ on Γ :

$$\mathcal{L} = \kappa^2 - \nabla \cdot (a \nabla),$$

with Kirchhoff vertex conditions

Fractional Differential Operators on Γ

Assume:

- $\kappa \in L^\infty(\Gamma)$: $\text{ess inf}_{x \in \Gamma} \kappa(x) \geq c > 0$
- $a : \Gamma \rightarrow \mathbb{R}$ is a strictly positive continuous function

Then, $\mathcal{L} = \kappa^2 - \nabla \cdot (a \nabla)$ is self-adjoint and positive definite, has a compact inverse, and thus admits a discrete spectrum $\{(\lambda_j, \varphi_j)\}_{j \in \mathbb{N}}$ ($0 < \lambda_1 \leq \lambda_2 \leq \dots$).

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The fractional power $\mathcal{L}^\beta$ can be defined spectrally

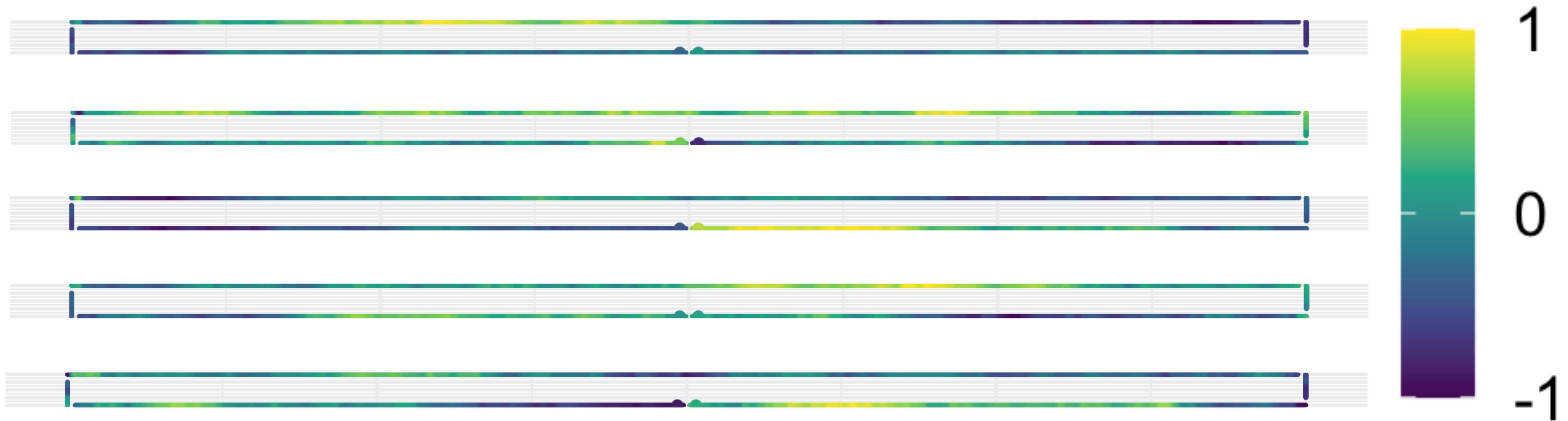
$$\mathcal{L}^\beta p = \sum_{j=1}^{\infty} \lambda_j^\beta \langle p, \varphi_j \rangle_{L^2(\Gamma)} \varphi_j, \quad p \in \mathcal{D}(\mathcal{L}^\beta).$$

Whittle-Matérn GP on metric graph Γ

Defined via a fractional SPDE: $(\kappa_{\text{ref}}^2 - \nabla \cdot (a_{\text{ref}} \nabla))^{\alpha} u = \mathcal{W}$ on Γ , with **standard Kirchhoff vertex conditions**.

- **White noise:** $\mathcal{W} = \sum_{k \in \mathbb{N}} \xi_k e_k$, where $\{\xi_k\}_{k \in \mathbb{N}}$ are i.i.d. $\mathcal{N}(0,1)$, $\{e_k\}_{k \in \mathbb{N}}$ is an orthonormal basis for $L^2(\Gamma)$. [**Bolin et al., *Math.Comp.*, 2024**]

Whittle-Matérn GP on Γ : Sample Paths



Five realizations of a GP on Γ

Metric Graph Setting

Choice of kernel: Whittle-Matérn GP on metric graph Γ

$$k(x, x') := \tau^{-2} \sum_{i=1}^{\infty} \lambda_i^{-2\alpha} \psi_i(x) \psi_i(x'), \quad x, x' \in \Gamma$$

with corresponding **RKHS**

$$\mathcal{H}_k = \left\{ g = \sum_{i=1}^{\infty} a_i \psi_i \ : \ \|g\|_{\mathcal{H}_k}^2 := \tau^2 \sum_{i=1}^{\infty} a_i^2 \lambda_i^{2\alpha} < \infty \right\} =: H^{2\alpha}(\Gamma).$$

Regret Bounds

Simple regret: $r_t^{alg} := f^\dagger(x^\star) - f^\dagger(x_t^\star)$, $x^\star = \arg \max_{x \in \Gamma} f^\dagger(x)$, $x_t^\star = \arg \max_{x \in \{x_i\}_{i=1}^t} f^\dagger(x)$

Theorem (Fully specified setting)

If $f^\dagger \in H^{2\alpha}(\Gamma)$ with $\alpha > 1/2$ then w.p. $1 - \delta$

$$r_T^{UCB} = O \left(T^{\frac{1-2\alpha}{4\alpha}} \log T + T^{\frac{1-4\alpha}{4\alpha}} \sqrt{\log T} \left(\|f^\dagger\|_{H^{2\alpha}(\Gamma)} + \sqrt{\log(1/\delta)} \right) \right)$$

$$r_T^{TS} = O \left(\left[T^{\frac{1-2\alpha}{4\alpha}} \log T + T^{\frac{1-4\alpha}{4\alpha}} \sqrt{\log T} \left(\|f^\dagger\|_{H^{2\alpha}(\Gamma)} + \sqrt{\log(1/\delta)} \right) \right] \log(\|f^\dagger\|_{H^{2\alpha}(\Gamma)}^2 T^6) \right)$$

Regret Bounds

We also obtain regret bounds in **misspecified** settings:

Epistemic: the **smoothness** of the objective and the kernel **do not match**.

Computational: Exact graph kernels require **eigenpairs** of the operator on Γ , which are **typically unavailable** for general metric graphs. In practice, we use **FEM discretizations** to approximate these kernels.

Misspecified Setting

Finite element kernel representation:

Let V_h be the continuous piecewise-linear FEM space on the metric graph:

$$V_h = \text{span}\{e_{h,i}\}_{i=1}^{N_h}, \quad N_h \simeq h^{-1}.$$

Let L_h be the FEM discretization of

$$\mathcal{L} = \kappa^2 - \Delta_\Gamma, \quad L_h = \kappa^2 I + C^{-1}G,$$

where $C_{i,j} = \langle e_{h,i}, e_{h,j} \rangle_{L^2(\Gamma)}$ and $G_{i,j} = \langle \nabla e_{h,i}, \nabla e_{h,j} \rangle_{L^2(\Gamma)}$.

Using FEM eigenpairs $\{(\lambda_{h,i}, \psi_{h,i})\}_{i=1}^{N_h}$, we define the finite element approximation kernel by:

$$k_{h,\alpha}(x, x') = \tau^{-2} \sum_{i=1}^{N_h} \lambda_{h,i}^{-2\alpha} \psi_{h,i}(x) \psi_{h,i}(x').$$

— defined spectrally through FEM eigenpairs

Misspecified Setting

If $2\alpha \in \mathbb{N}$:

The kernel can be assembled directly from the FEM mass and stiffness matrices.

$$k_{h,\alpha}(x, x') = \tau^{-2} e_h(x)^\top Q_{h,\alpha}^{-1} e_h(x'), \quad Q_{h,\alpha} = C(\kappa^2 I + C^{-1} G)^{2\alpha},$$

where $e_h(x) = (e_{h,1}(x), \dots, e_{h,N_h}(x))$.

Misspecified Setting

If $2\alpha \notin \mathbb{N}$: following the rational SPDE approach of Bolin--Kirchner [2020]

we approximate the fractional power on the FEM spectrum by a rational function s_h :

$$s_h(z) \approx z^\alpha, \quad z \in J_h := [\lambda_{h,N_h}^{-1}, \lambda_{h,1}^{-1}].$$

Thus $s_h(\lambda_{h,i}^{-1}) \approx \lambda_{h,i}^{-\alpha}$, which gives the rational FEM kernel

$$k_{h,\alpha}^r(x, x') = \tau^{-2} \sum_{i=1}^{N_h} s_h(\lambda_{h,i}^{-1})^2 \psi_{h,i}(x) \psi_{h,i}(x').$$

Using $s_h(z^{-1}) = \frac{p_r(z)}{p_l(z)}$, we can express the rational approximation $k_{h,\alpha}^r$ **without explicitly computing the spectral decomposition**:

$$k_{h,\alpha}^r(x, x') = \tau^{-2} e_h(x)^\top P_r (P_l^\top C P_l)^{-1} P_r^\top e_h(x'),$$

where $P_r := p_r(\kappa^2 I + C^{-1}G)$, and $P_l := p_l(\kappa^2 I + C^{-1}G)$.

Regret Bounds

Simple regret: $r_t^{alg} := f^\dagger(x^\star) - f^\dagger(x_t^\star)$, $x^\star = \arg \max_{x \in \Gamma} f^\dagger(x)$, $x_t^\star = \arg \max_{x \in \{x_i\}_{i=1}^t} f^\dagger(x)$

Theorem (Misspecified setting)

If $f^\dagger \in H^{2\beta}(\Gamma)$ with $\beta > 1/2$ and kernel $k_{h,\alpha}$ ($k_{h,\alpha}^r$) with $\alpha > 1/2$ and mesh-size h , then:

$$r_T^{UCB} = O \left(T^{\frac{-(4\alpha-3)}{8\alpha-2}} \log T + T^{\frac{-(2\alpha-1)}{4\alpha-1}} h^{(2\beta-2\alpha) \wedge 0} \sqrt{\log T} + h^{(2\beta-1) \wedge \frac{3}{2}} T^{\frac{1}{8\alpha-2}} \right)$$

$$r_T^{TS} = O \left(\left[T^{\frac{-(4\alpha-3)}{8\alpha-2}} \log T + T^{\frac{-(2\alpha-1)}{4\alpha-1}} h^{(2\beta-2\alpha) \wedge 0} \sqrt{\log T} + h^{(2\beta-1) \wedge \frac{3}{2}} T^{\frac{1}{8\alpha-2}} \right] \log(T^2/h) \right)$$

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Experimental Setting

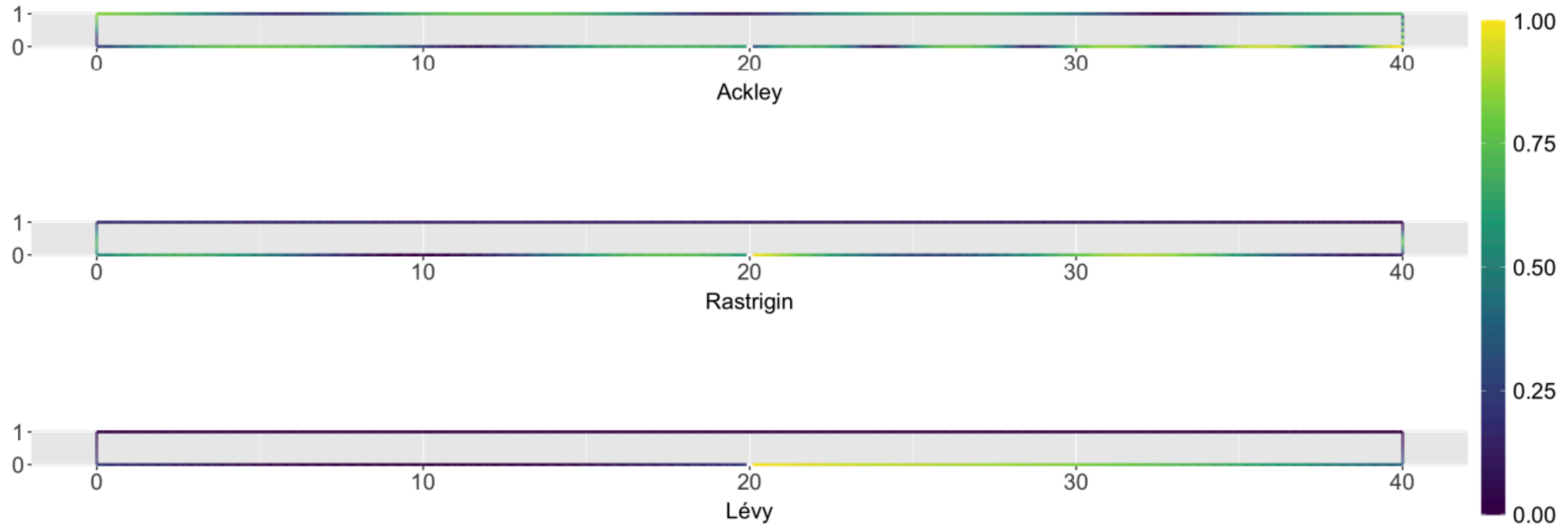
We compare two kernels:

- **SPDE Whittle-Matérn kernel** on metric graph with $\alpha = 1$
- **Euclidean kernel** $k_{\text{Eucl}}(x, x') = \sigma^2 \exp\left(-\frac{|x - x'|}{\ell}\right), \quad \ell > 0, \sigma > 0$

Performance metrics:

- Simple regret
- Reach rate
- Iterations to Tol

Benchmark Functions

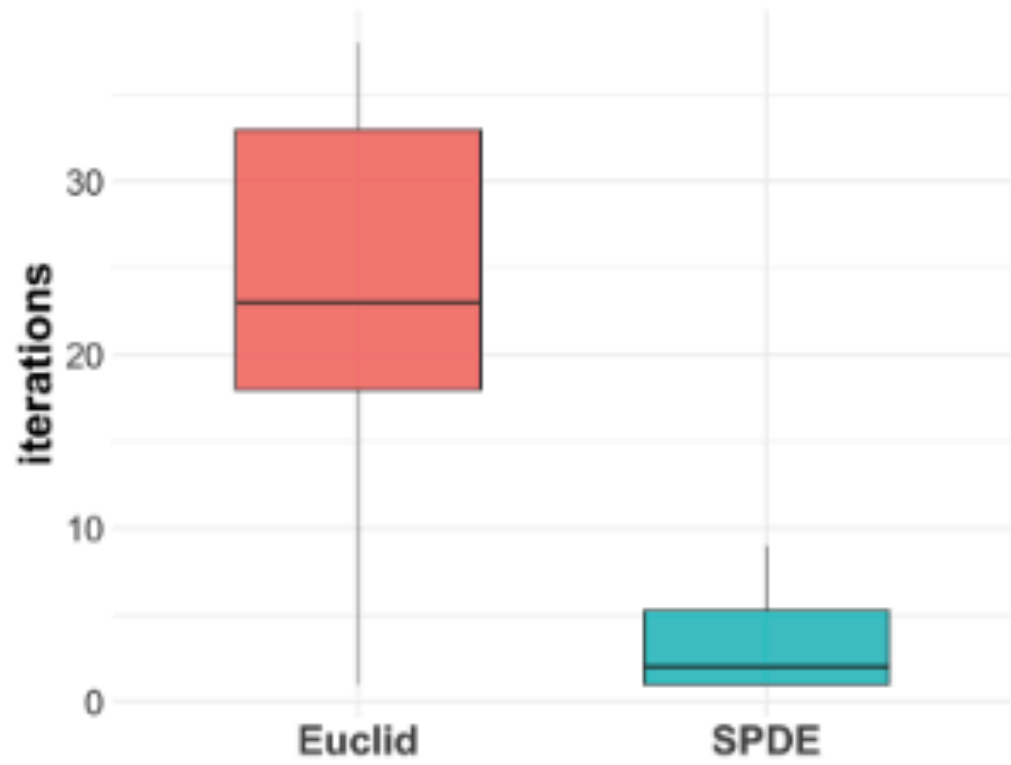
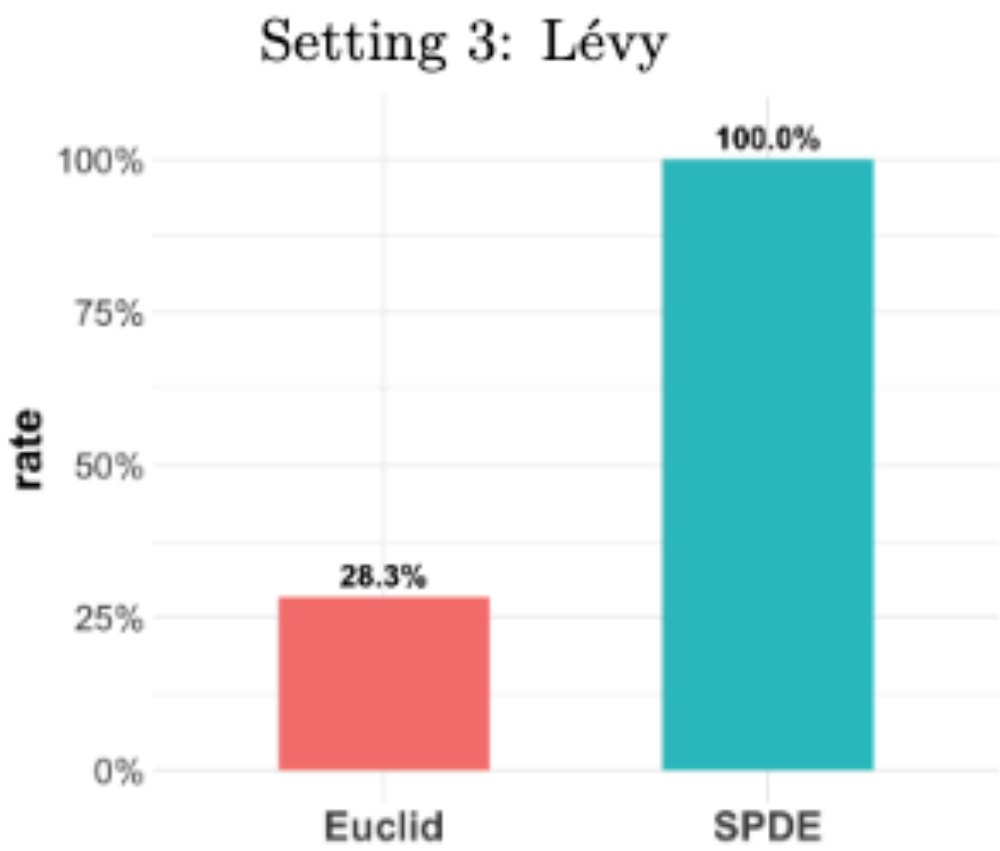
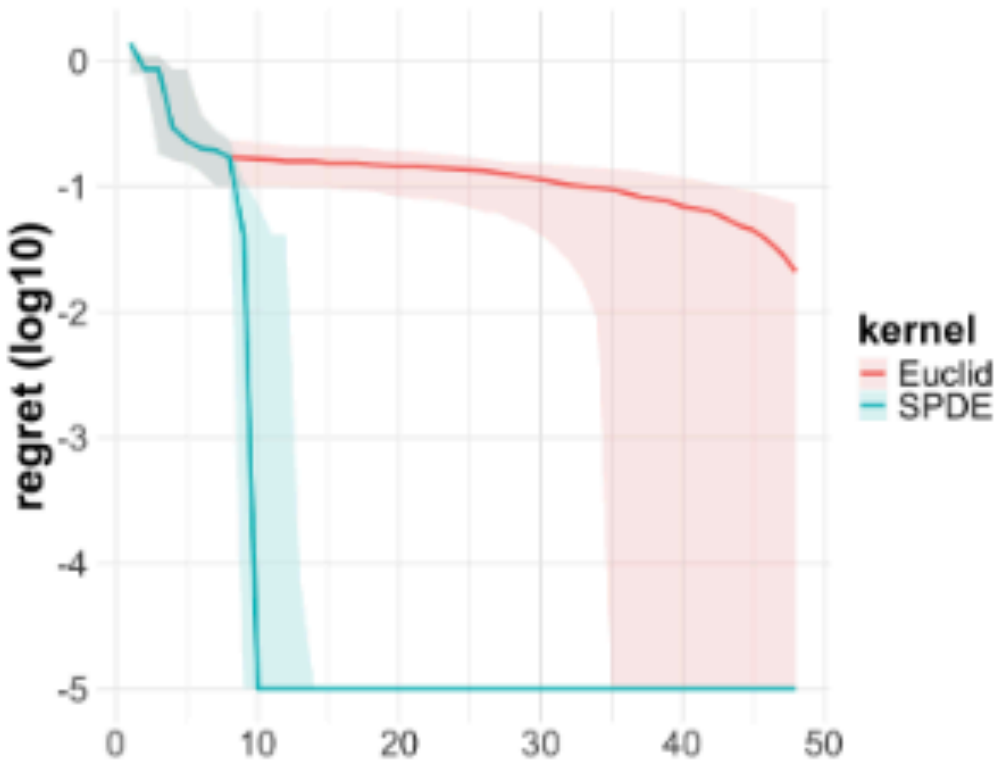
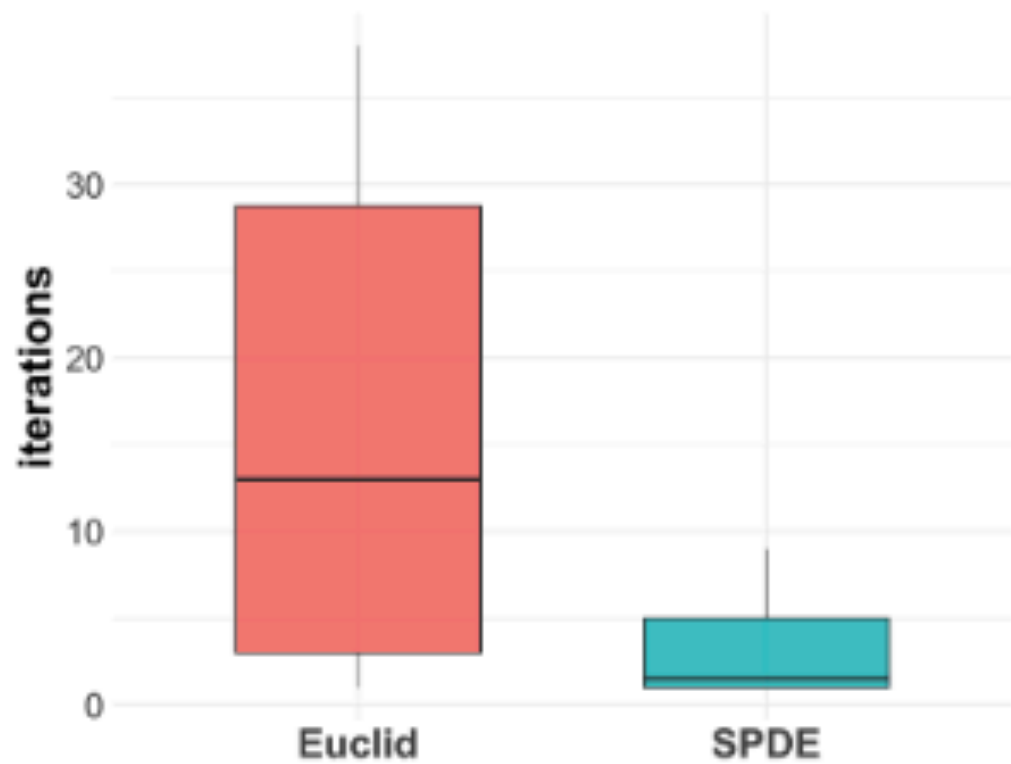
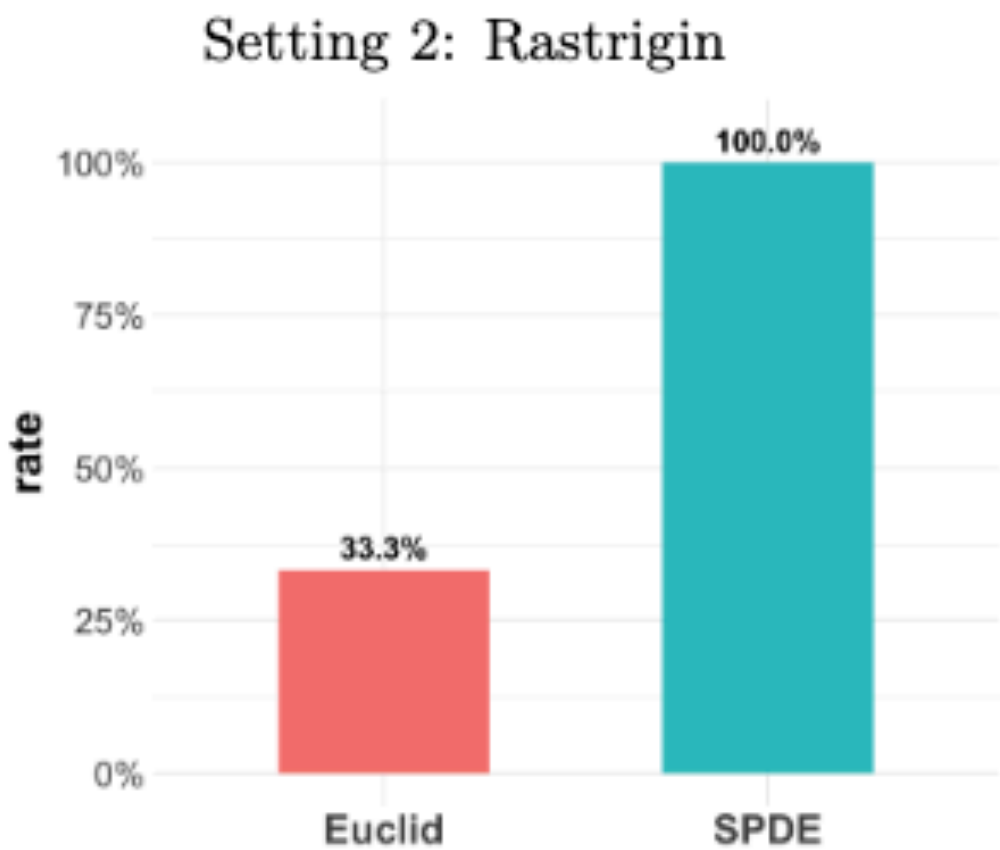
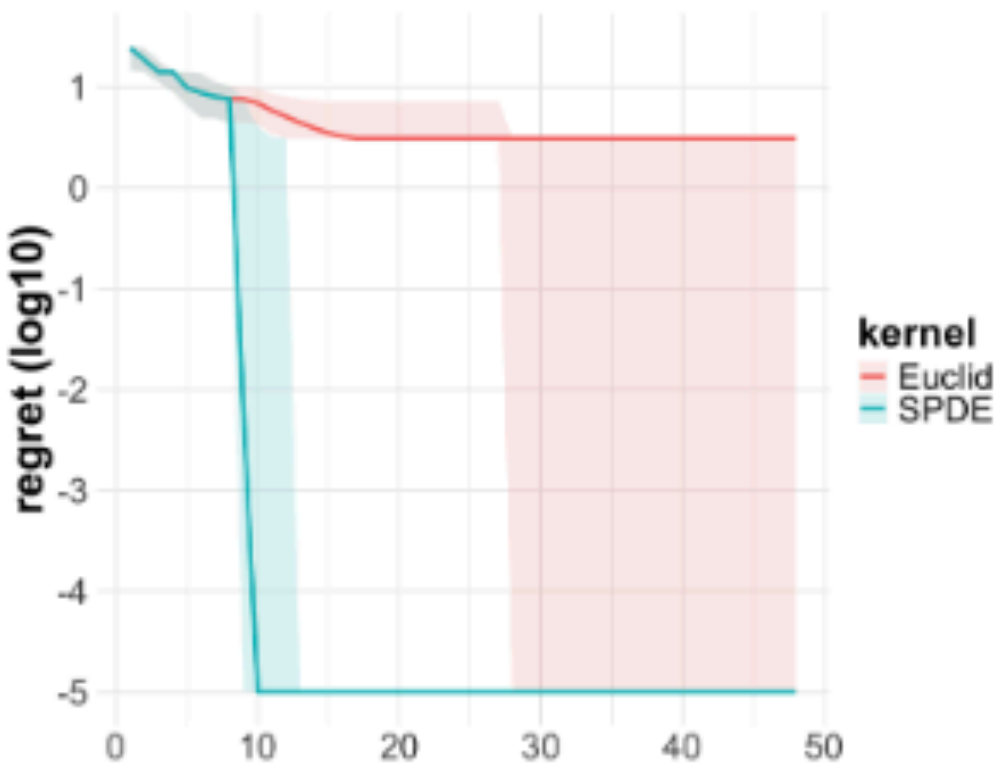
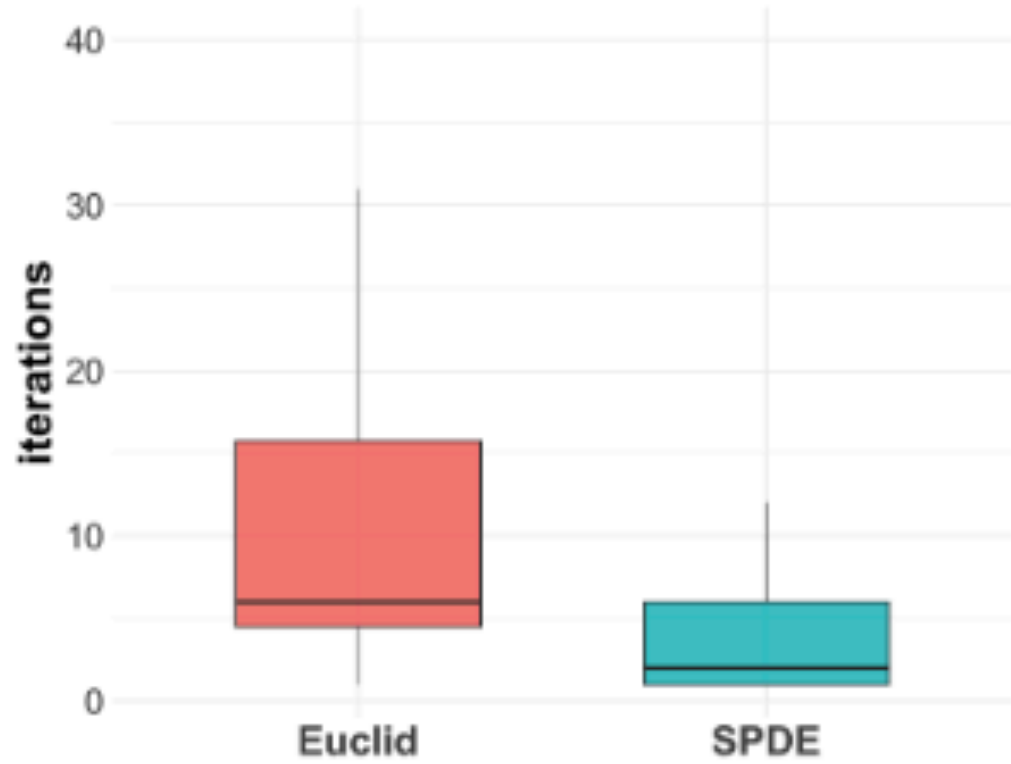
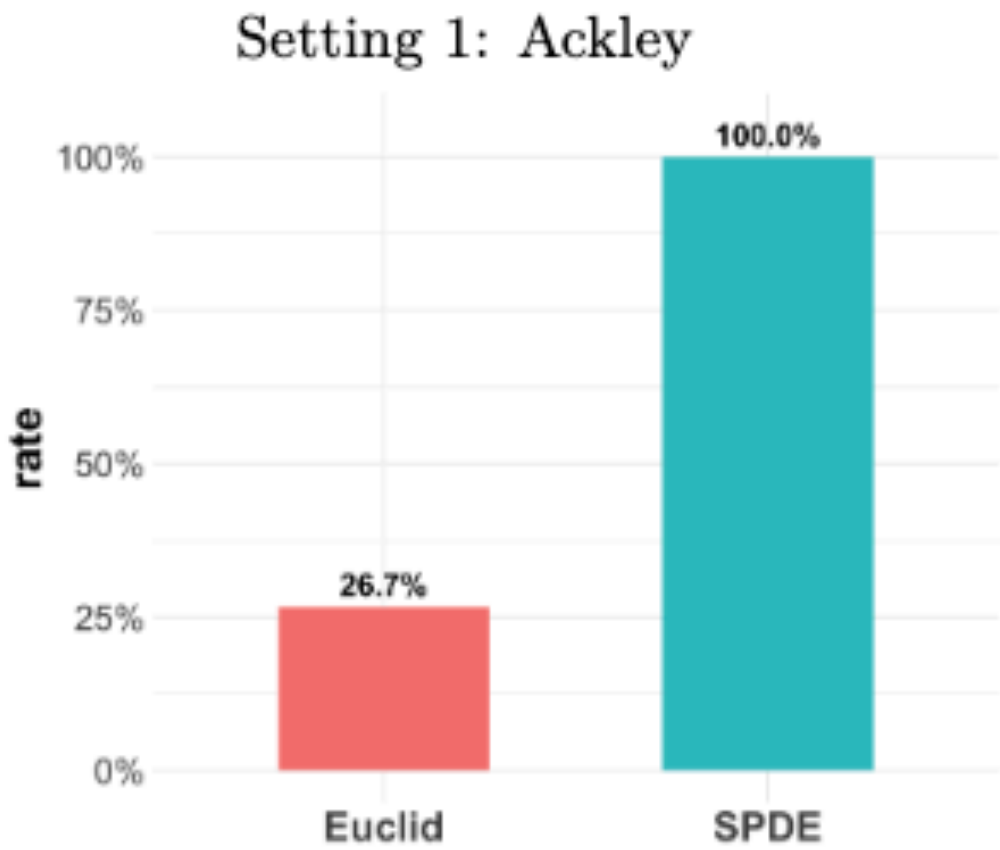
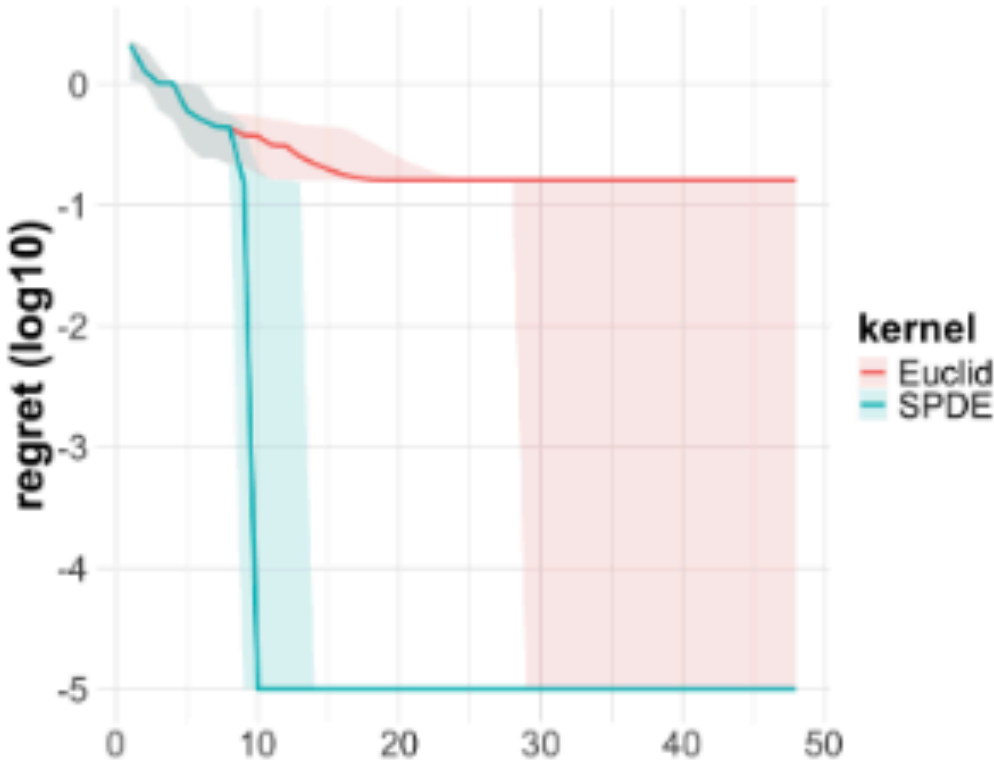


Ackley (1D): $g_{Ack}(x) = -20 \exp(-0.2 |x|) - \exp(\cos(2\pi x)) + 20 + e,$

Rastrigin (1D): $g_{Ras}(x) = x^2 - 10 \cos(2\pi x) + 10,$

Lévy (1D): $g_{Lev}(x) = \sin^2(\pi w) + (w - 1)^2 \left[1 + \sin^2(2\pi w) \right], \quad w := 1 + \frac{x - 1}{4}.$

IGP-UCB

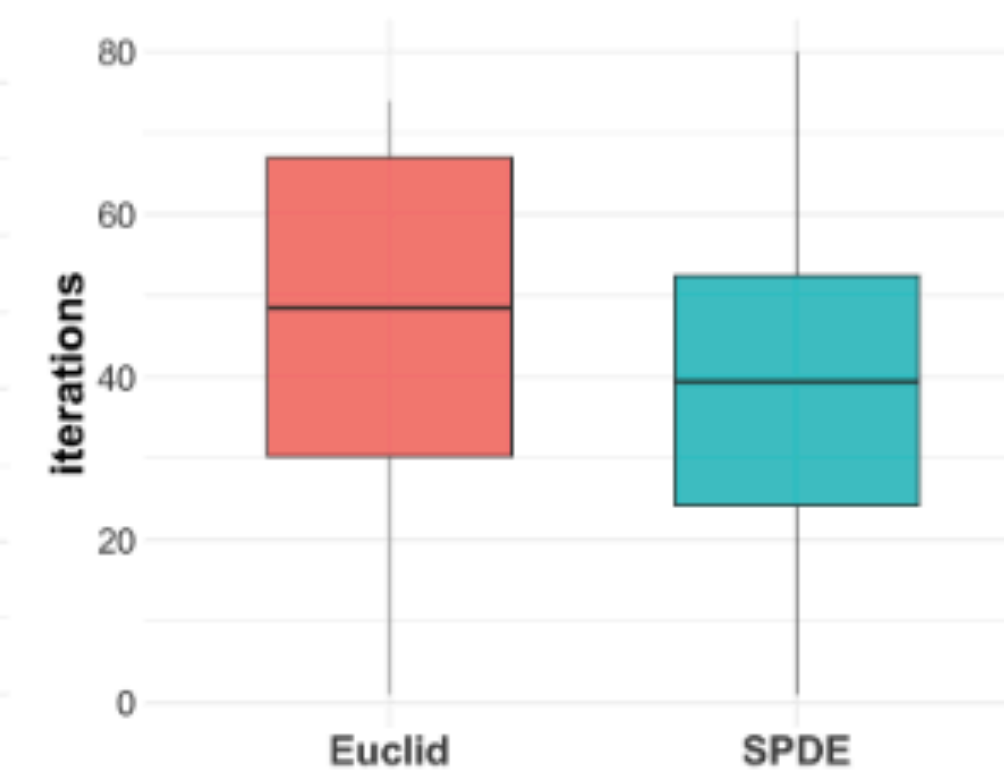
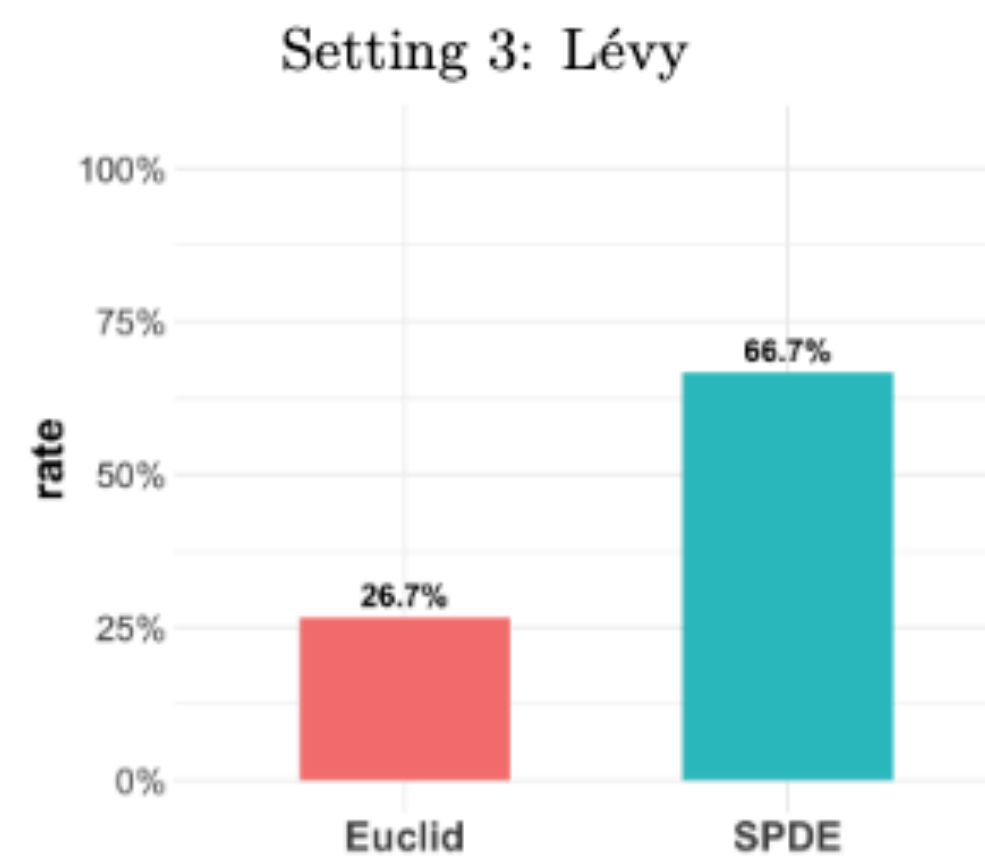
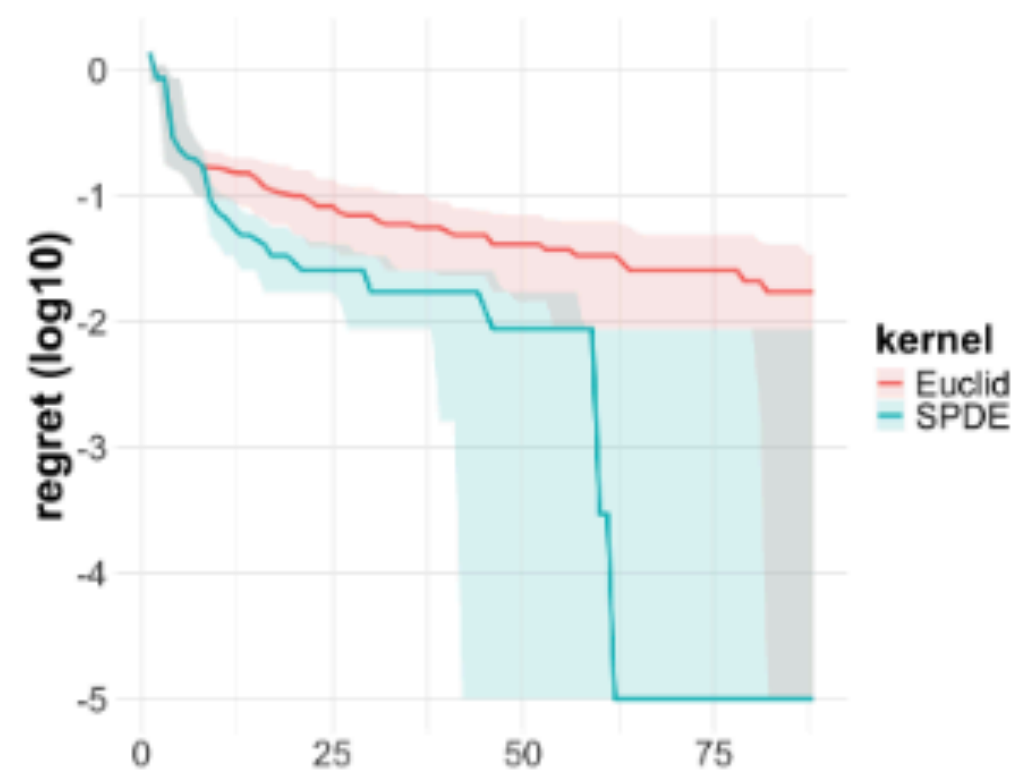
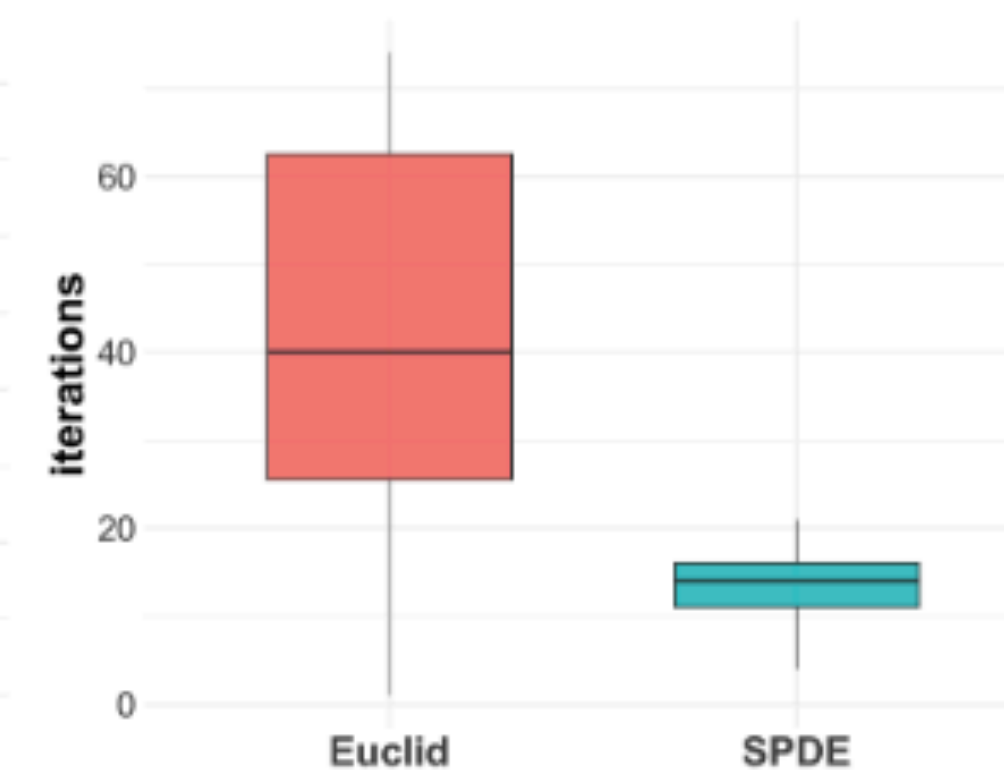
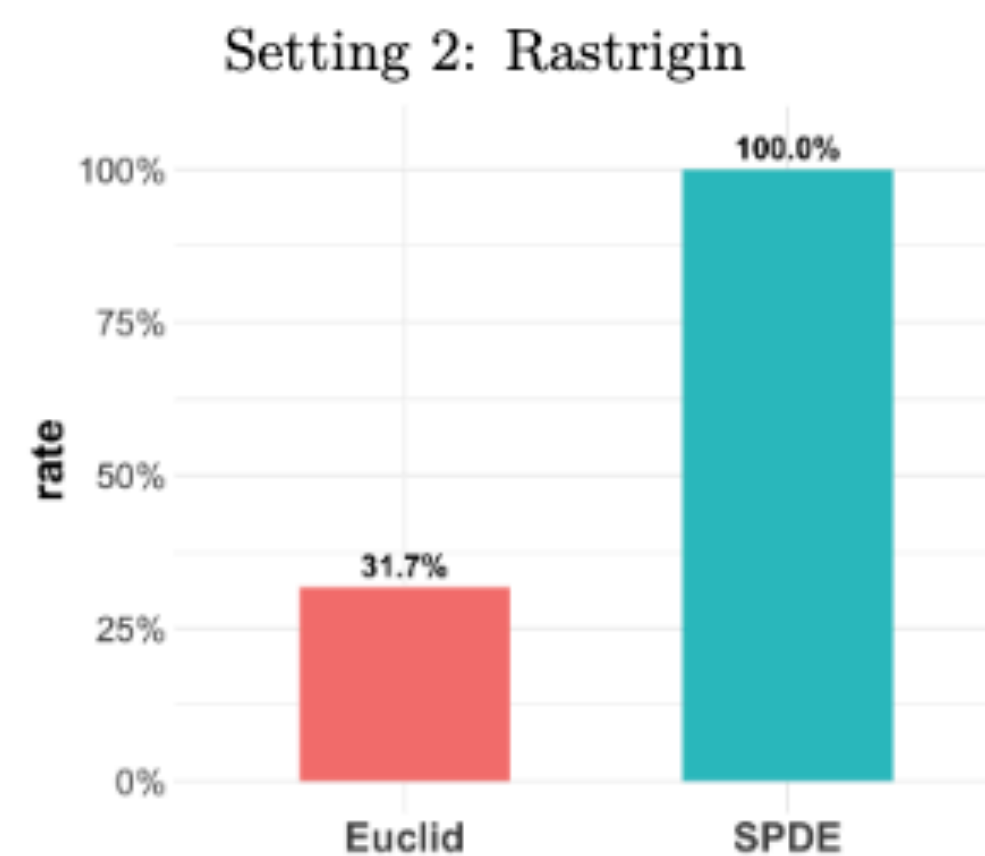
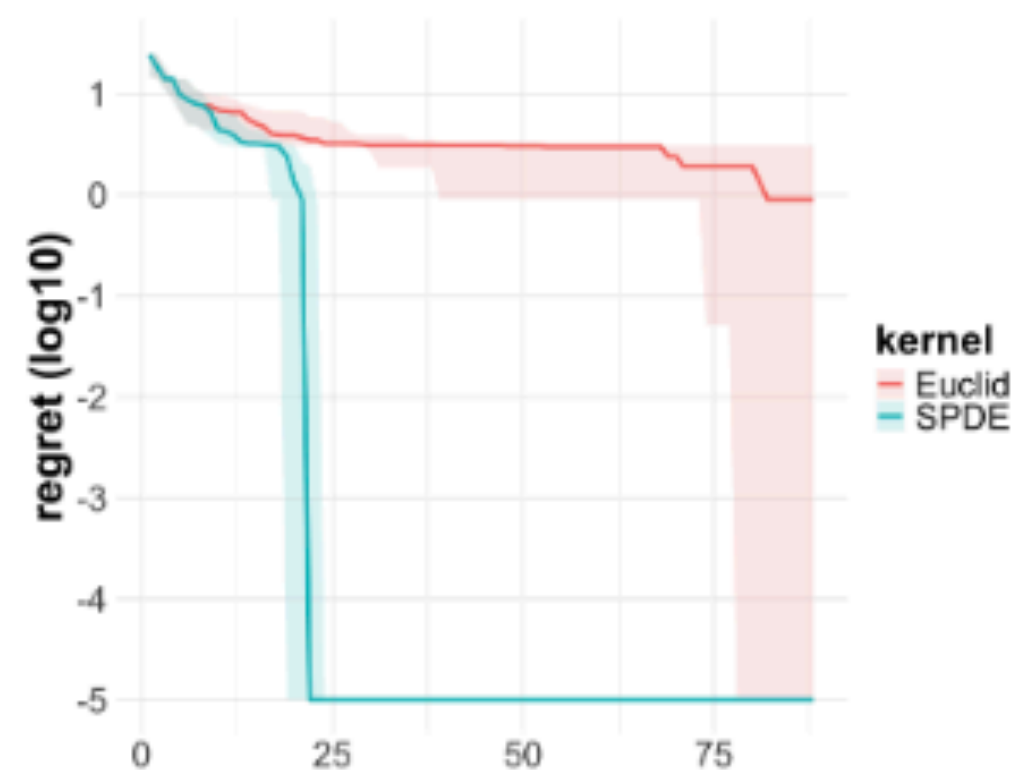
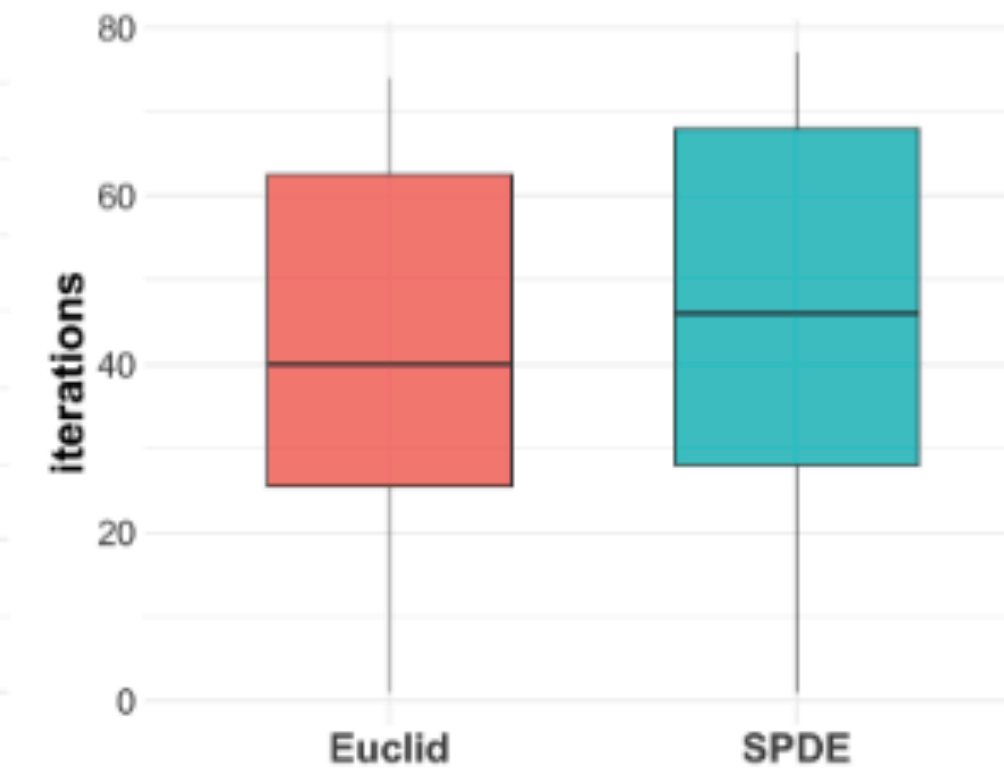
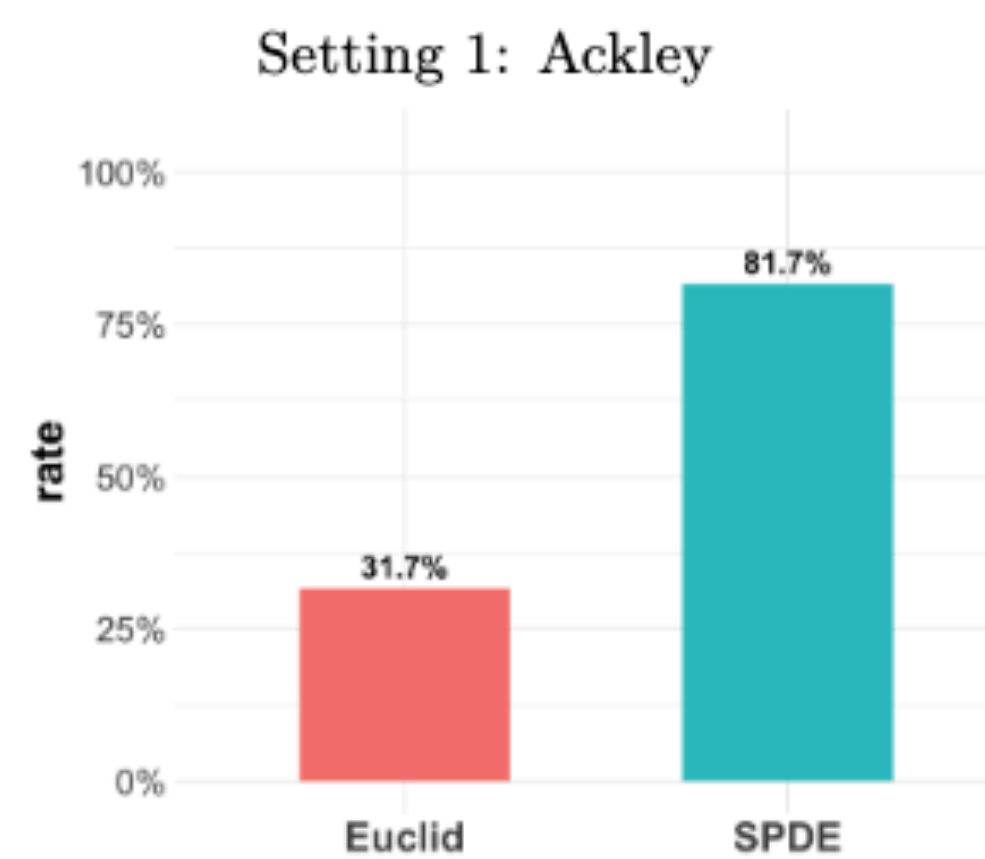
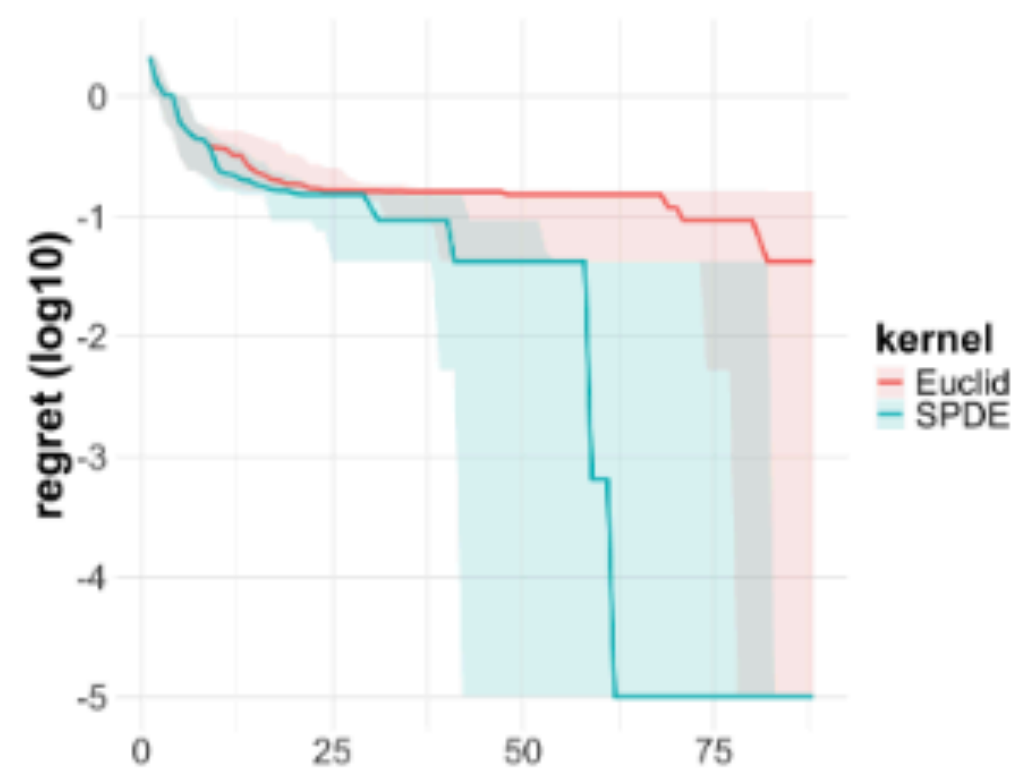


(a) Simple regret

(b) Reach rate

(c) Iterations to Tol

GP-TS



(a) Simple regret

(b) Reach rate

(c) Iterations to Tol

MAP Estimation for Bayesian Inversion

Inverse problem:

On a compact metric graph Γ , we can only access to the noisy observations of the solution to an elliptic PDE $(\chi^2 - \Delta_\Gamma)p = g$,

where $\chi > 0$, and g is a source function localized around an **unknown point** $x^\dagger \in \Gamma$.

Goal: Infer the source location $x^\dagger$.

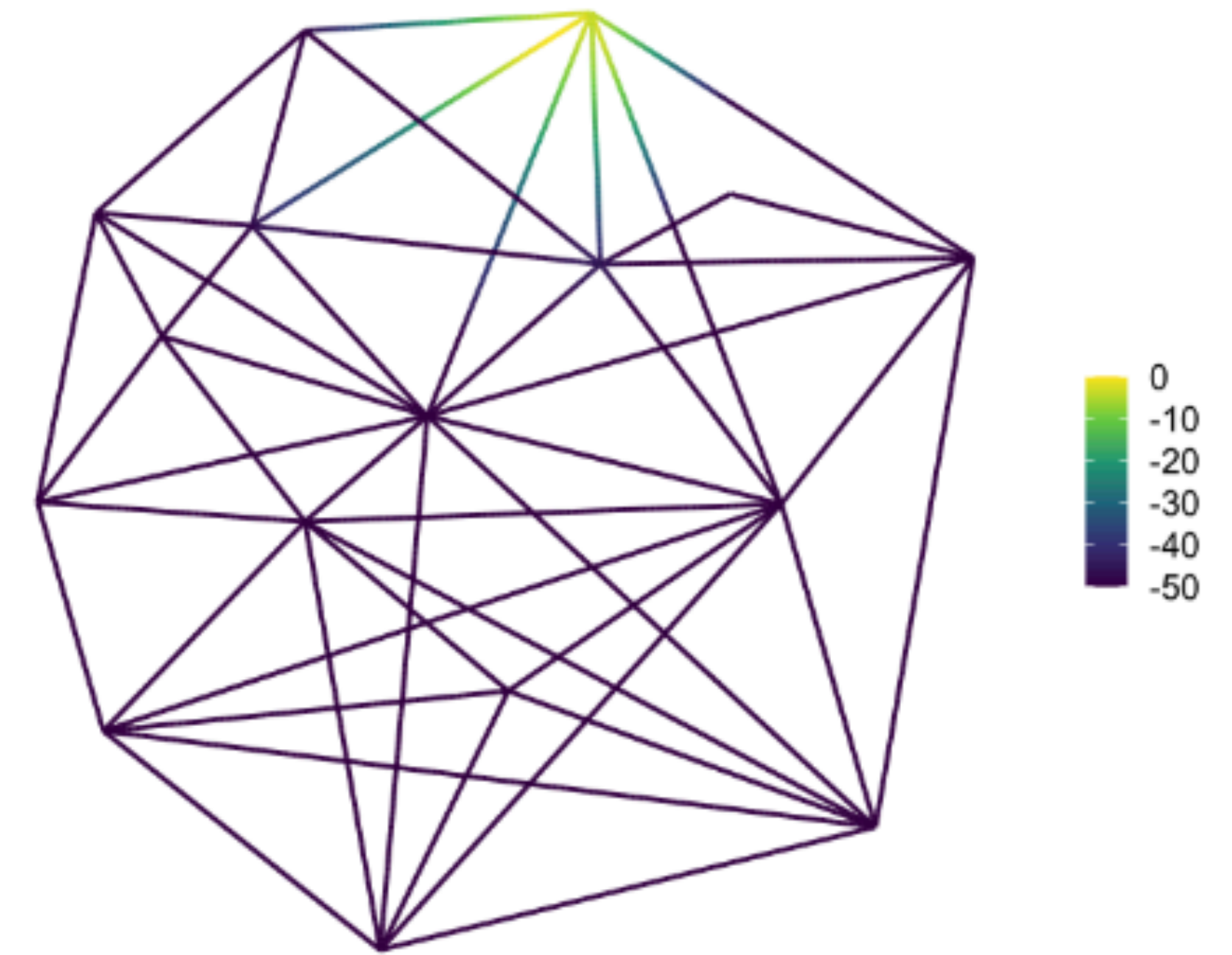
MAP Estimation for Bayesian Inversion

Objective (log-posterior)

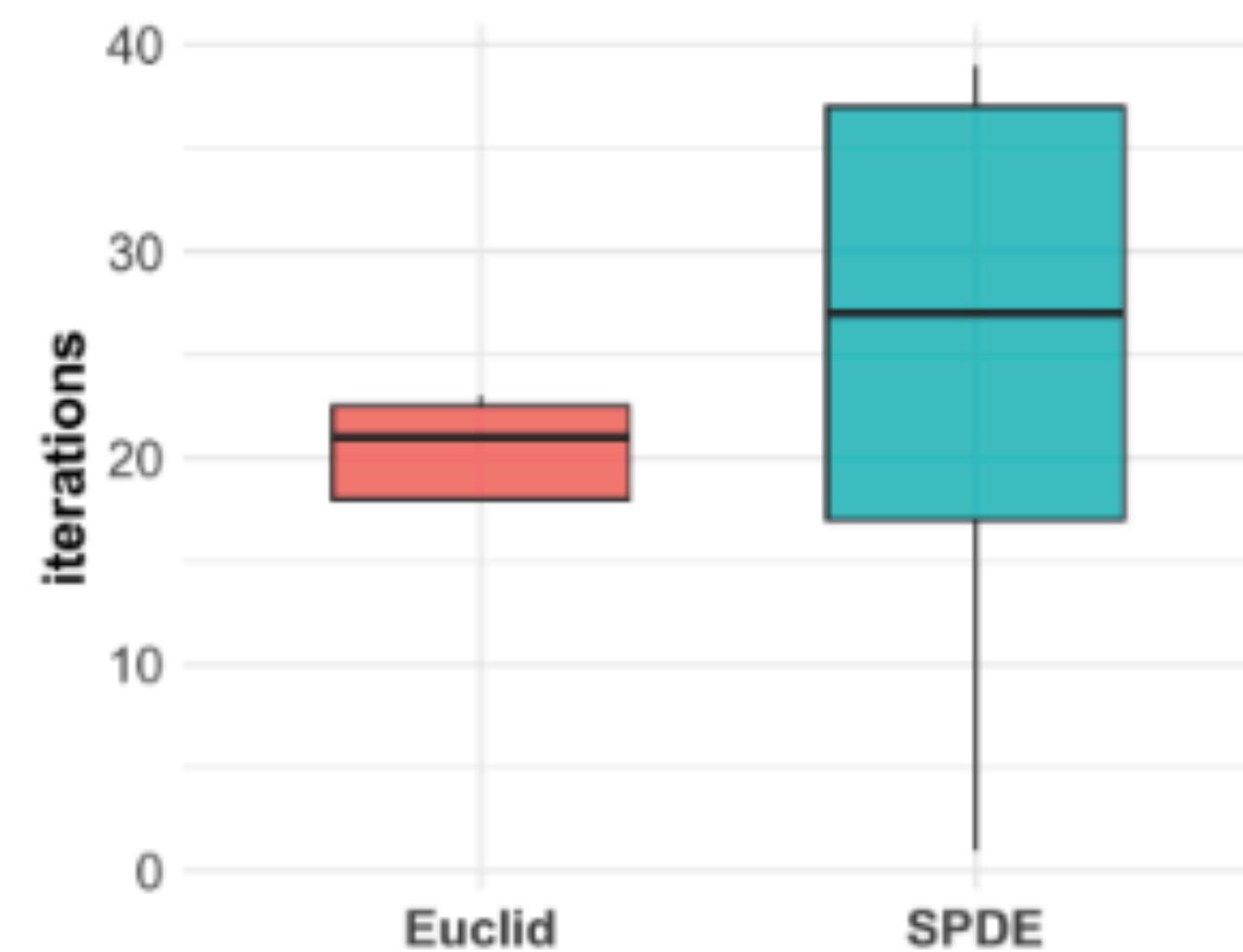
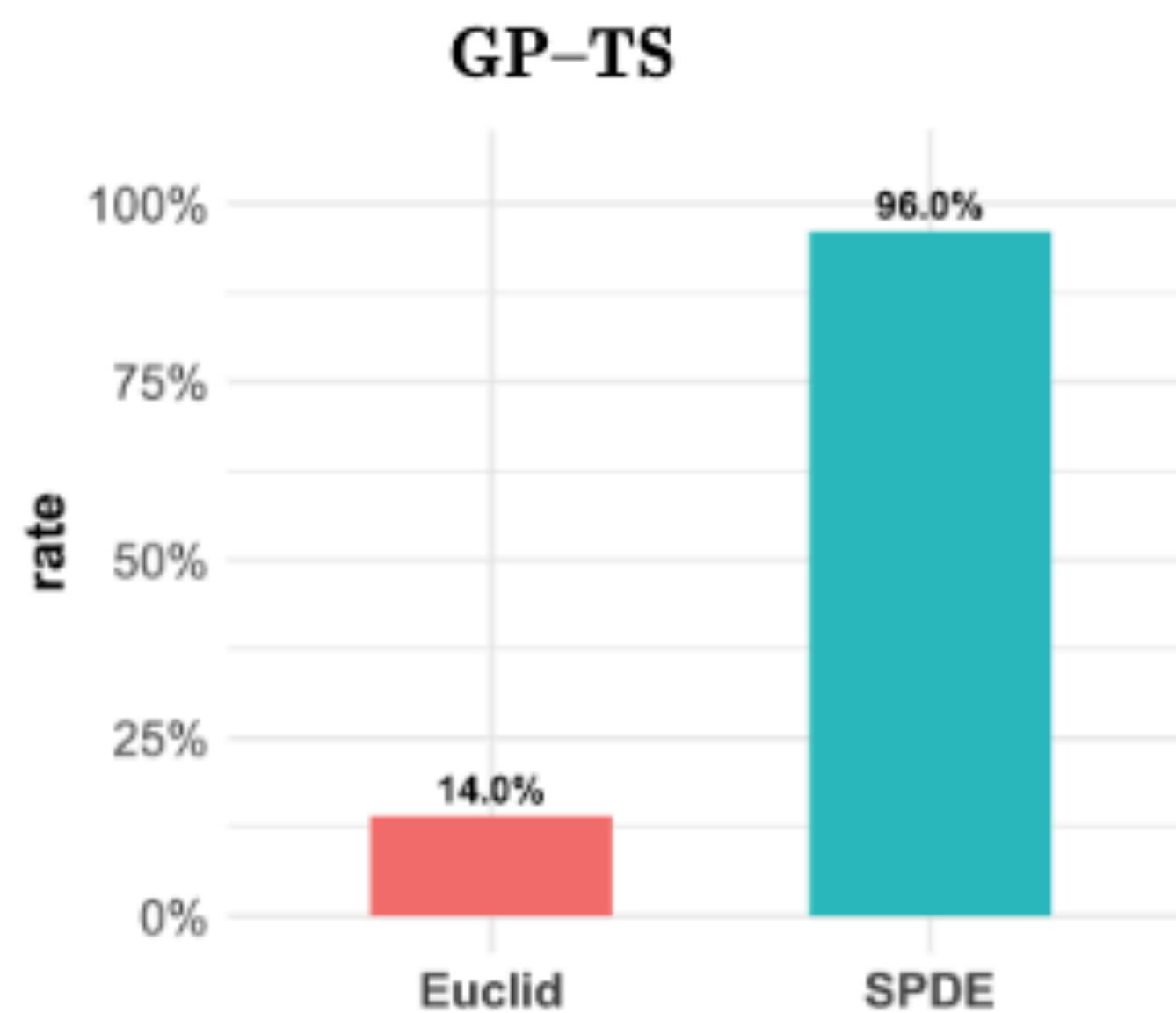
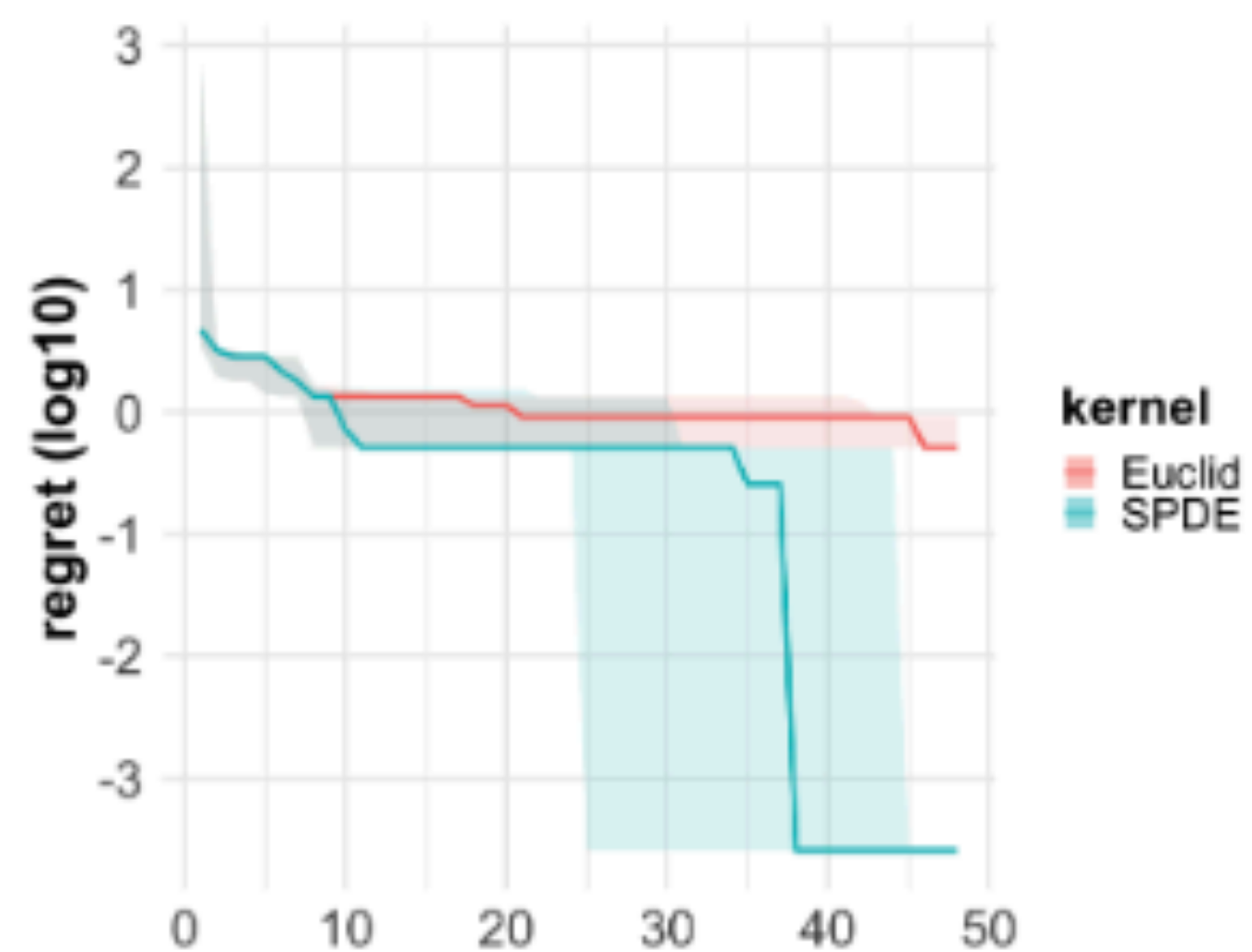
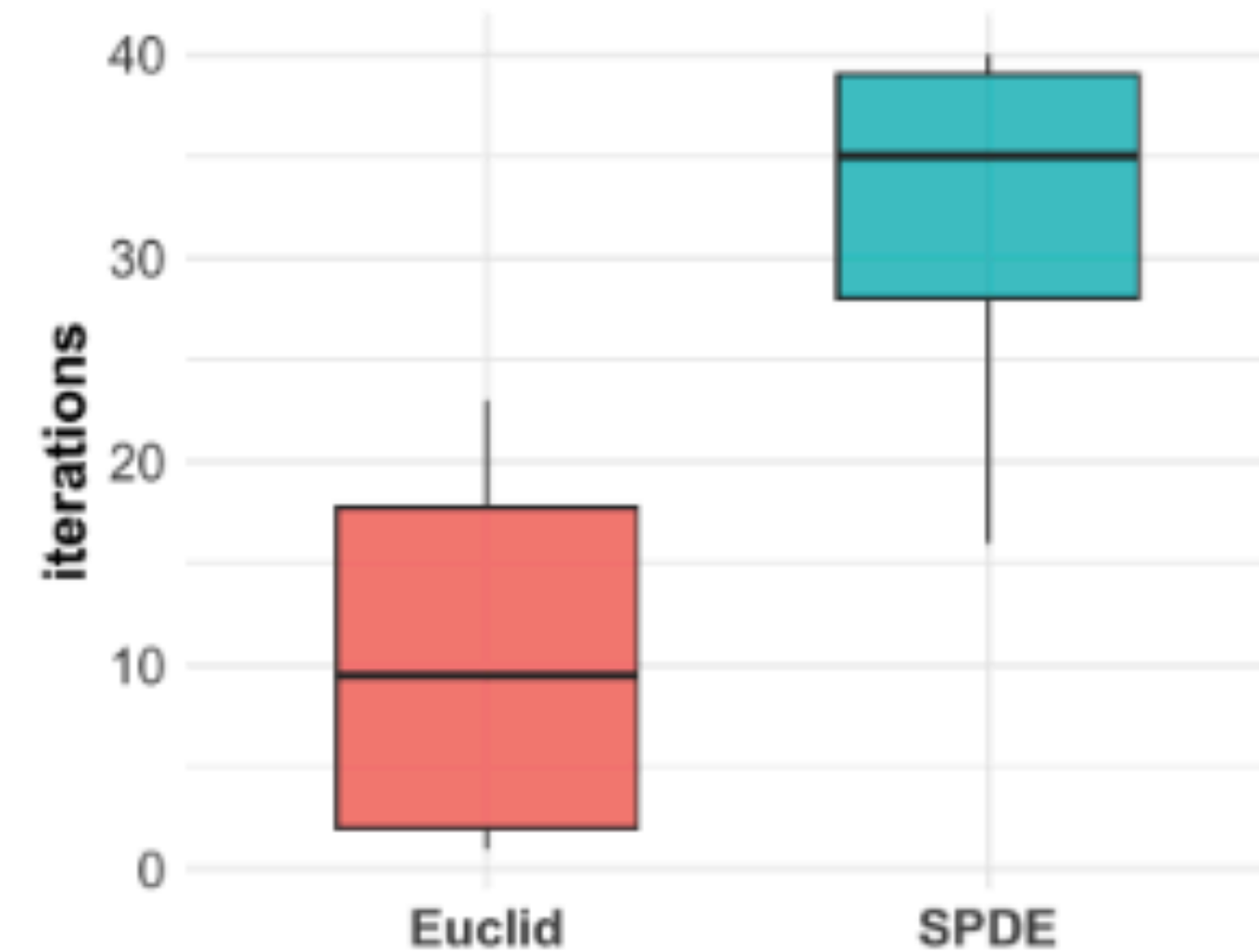
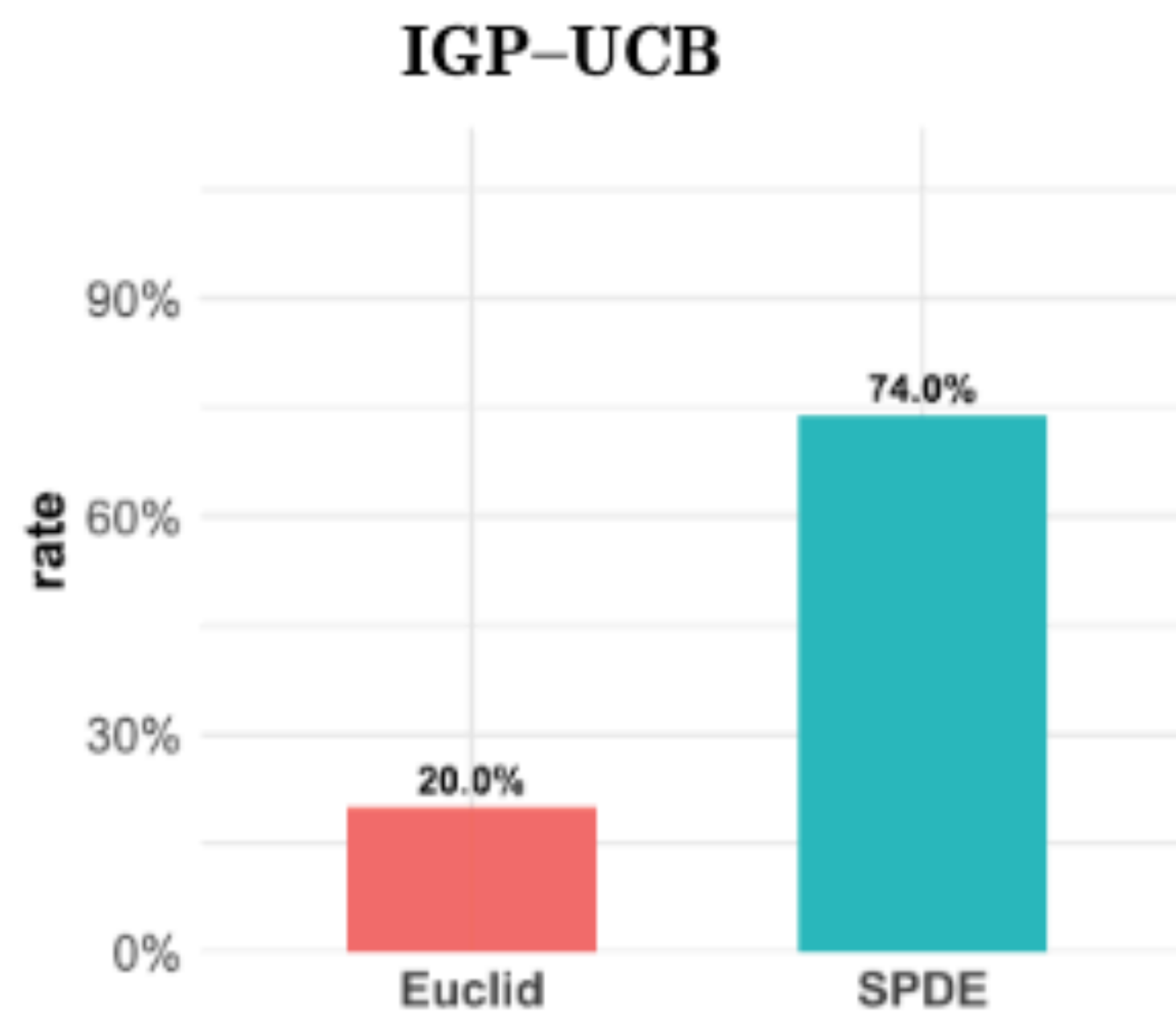
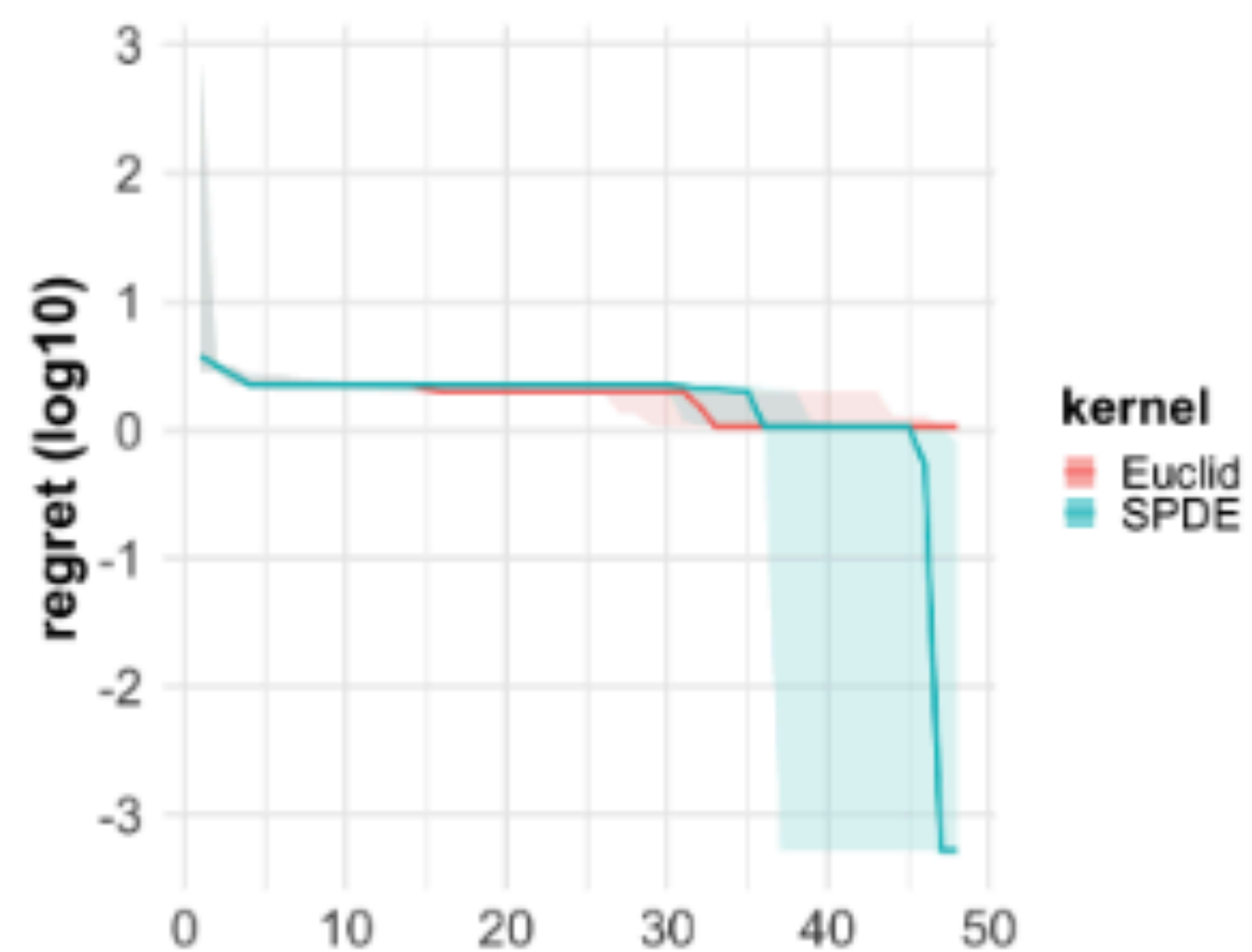
$$LP(x) = -\frac{1}{2} \|D - \mathcal{G}(x)\|^2 + \log \pi(x)$$

$$x_{\text{MAP}} = \arg \max_{x \in \Gamma} LP(x).$$

- x : candidate source location on the graph
- $\pi(x)$: prior density
- $\mathcal{G}(x)$: forward model from source location to observations of PDE solution p
- D : observed data (noisy observations of PDE solution p)



(b) Log-posterior $LP(x)$ for source location



(a) Simple regret

(b) Reach rate

(c) Iterations to Tol

Summary

- We have introduced Bayesian optimization on metric graphs
- Kernels adapted to intrinsic geometry crucial in theory and practice
- Regret bounds in specified and misspecified settings
- Numerical results for benchmark objectives and MAP estimation

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Journal of Computational Physics, 2026

References

Bayesian optimization on networks

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